

$$1. \quad \beta = 61.9 \text{ dB} \therefore v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}, I = \frac{(\Delta p)^2}{2 \rho v_w}, \beta (\text{dB}) = 10 \log_{10} \left(\frac{I}{I_0} \right), I_0 = 1 \times 10^{-12} \frac{\text{W}}{\text{m}^2}$$

$$2. \quad v_{\max} = 0.0737 \text{ m/s} \therefore v(t) = \sqrt{\frac{k}{m}} X \sin \frac{2\pi t}{T} \text{ and the max of sine's range is 1.}$$

$$3. \quad f = 1.49 \text{ s} \therefore T_{\text{SHO}} = 2\pi \sqrt{\frac{m}{k}}$$

$$4. \quad f = 4.55 \times 10^3 \text{ Hz} \therefore f_n = n \frac{v_w}{4L}, v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}$$

$$5. \quad k = 9.60 \times 10^4 \frac{\text{N}}{\text{m}} \therefore \vec{F} = -k \Delta \vec{x}. k = -\frac{\vec{F}}{\Delta \vec{x}}$$

$$6. \quad PE = 1.58 \text{ J} \therefore PE = \frac{1}{2} k x^2$$

$$7. \quad f = 3.70 \times 10^6 \text{ Hz} \therefore T = \frac{1}{f}$$

$$8. \quad d = 5.24 \times 10^3 \text{ m} \therefore \vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t}$$

$$9. \quad L = 0.2858 \text{ m} \therefore f_n = n \frac{v_w}{4L}, v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}$$

$$10. \quad L = 6.97 \text{ cm} \therefore T_{\text{pend}} = 2\pi \sqrt{\frac{L}{g}}$$

$$11. \quad a = 0.773 \% \therefore Z = \rho v, a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2}$$

$$12. \quad d = 1.26 \text{ m} \therefore \Delta E_t = W_{nc} = -f_k d, f_k = \mu_k N = \mu_k m g, PE = \frac{1}{2} k x^2$$

$$13. \quad f = 221.0 \text{ Hz} \therefore f_{\text{obs}} = f_s \left(\frac{v_w}{v_w \pm v_s} \right) \text{ (review the Doppler effect)}$$